



MATLAB implementation of a DPLL for QPSK modulation – Extension to 8PSK

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1 QPSK modulation : NDA PED

This section includes an analysis of Non-Data Aided (NDA) and Data-Directed (DD) Phase Error Detectors (PEDs) for QPSK modulation. It covers the construction of S-curves, phase ambiguity resolution, and closed-loop study involving acquisition and tracking phases.

1.1 S-curve

1.1.1 Matlab chain

I implemented the following chain on Matlab in purpose to draw the S-curve for QPSK modulation.

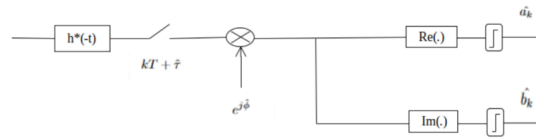


Figure 1: Open loop chain

1.1.2 Plot of S curve of the PED

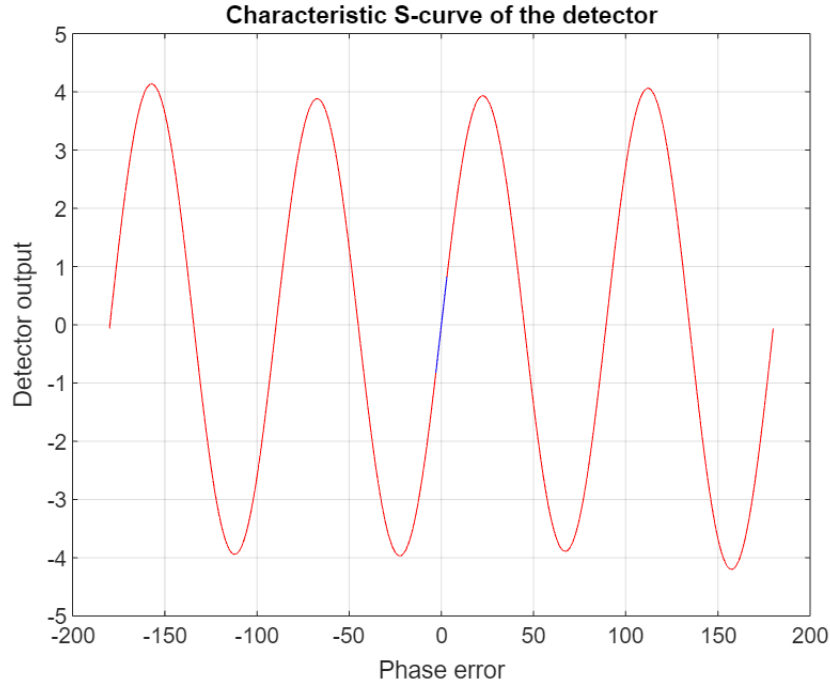


Figure 2: S-curve for a phase between -180° and 180°

1.1.3 Phase ambiguity

By analyzing the roots of the likelihood function's derivative, we identify the points that optimize the function. The resulting graph reveals multiple zeros, leading to a phase ambiguity of $\frac{\pi}{2}$.

It's crucial to navigate the estimated phase to sidestep points of unstable equilibrium and prevent cycle slipping. The adaptation algorithm is described as follows:

- When $\epsilon(m) = \theta - \hat{\theta}_m > 0$, indicating $\hat{\theta}_m < \theta$, we increase $\hat{\theta}_m$ towards θ .
- Conversely, if $\epsilon(m) = \theta - \hat{\theta}_m < 0$ (where $\hat{\theta}_m > \theta$), we decrease $\hat{\theta}_m$ to converge to θ .

Using the update :

$$\theta_{m+1} = \theta_m + L'(\theta)$$

1.2 Closed loop study

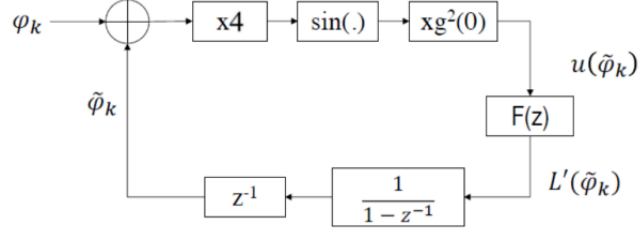


Figure 3: Structure of closed loop chain of QPSK modulation

1.2.1 Acquisition

The expression of the output of the detector is:

$$u_k(\theta - \tilde{\theta}) = -\Im \left(p(k, \hat{\tau}) \exp(-j\tilde{\theta}) \right)$$

and to implement that in Matlab i used this formula:

```
out_det(ii) = -imag((recu(ii)*exp(-1j*phi_est(ii)))^4)/slope;
```

We divide by the slope of the S-curve in the origin to improve the stability and accuracy of the DPLL.

Loop noise bandwidth is fixed on 0.01 and loop order is 2

1. DPLL response to a phase shift of 10° for $E_b/N_0 = 100$ dB

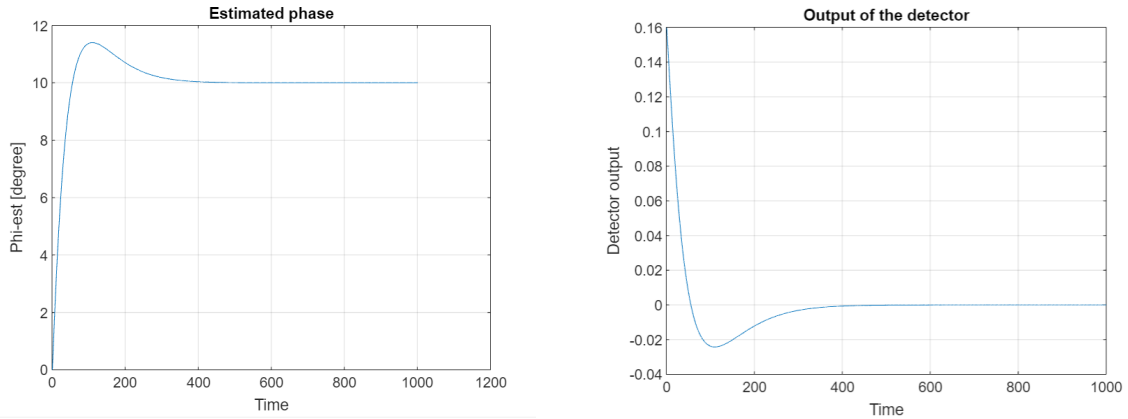


Figure 4: DPLL response to $\Delta\phi = 10^\circ$ for $\frac{E_b}{N_0} = 100$ dB

2. DPLL response to $\Delta\phi = 10^\circ$ for $E_b/N_0 = 7$ dB

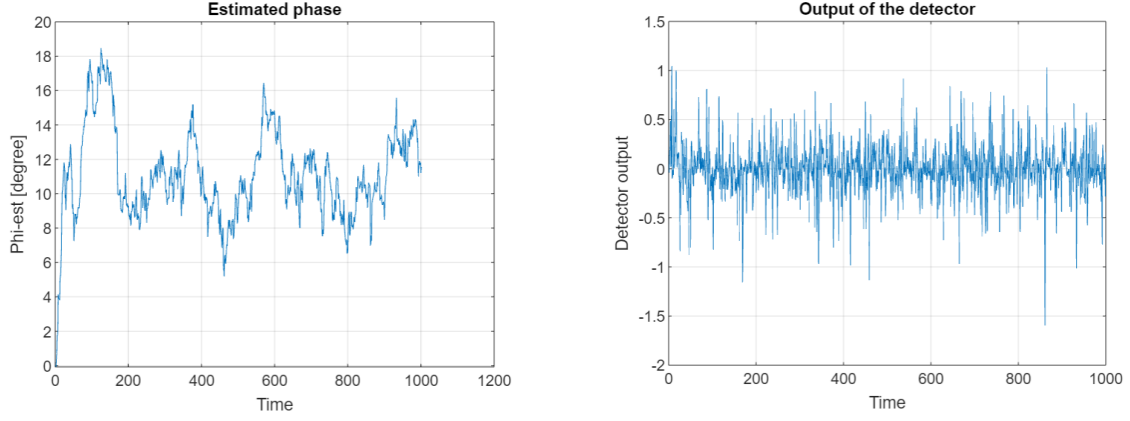


Figure 5: DPLL response to $\Delta\phi = 10^\circ$ for $E_b/N_0 = 7$ dB

3. DPLL response to $\frac{\delta f}{R_s} = 1\%$ for $\frac{E_b}{N_0} = 100$ dB

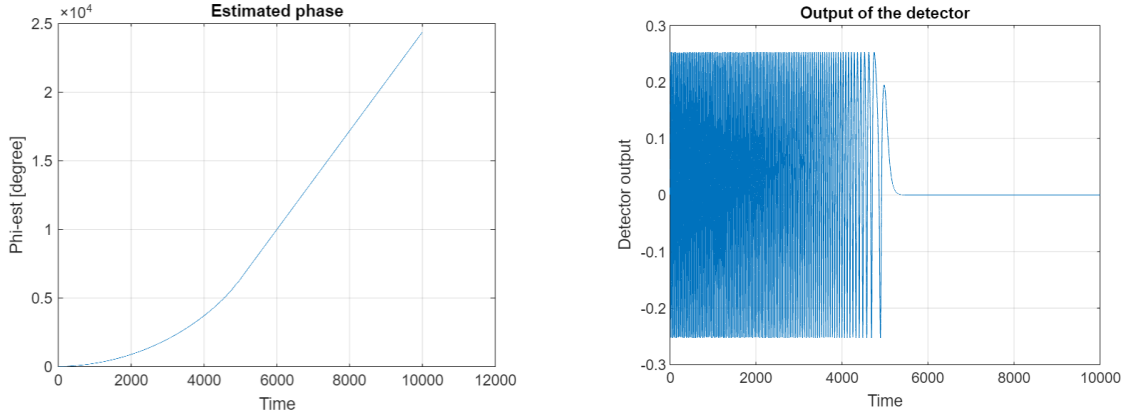


Figure 6: DPLL response to $\frac{\delta f}{R_s} = 1\%$; $\Delta\phi = 0^\circ$ for $E_b/N_0 = 100$ dB

4. DPLL response to $\frac{\delta f}{R_s} = 1\%$ for $\frac{E_b}{N_0} = 7dB$

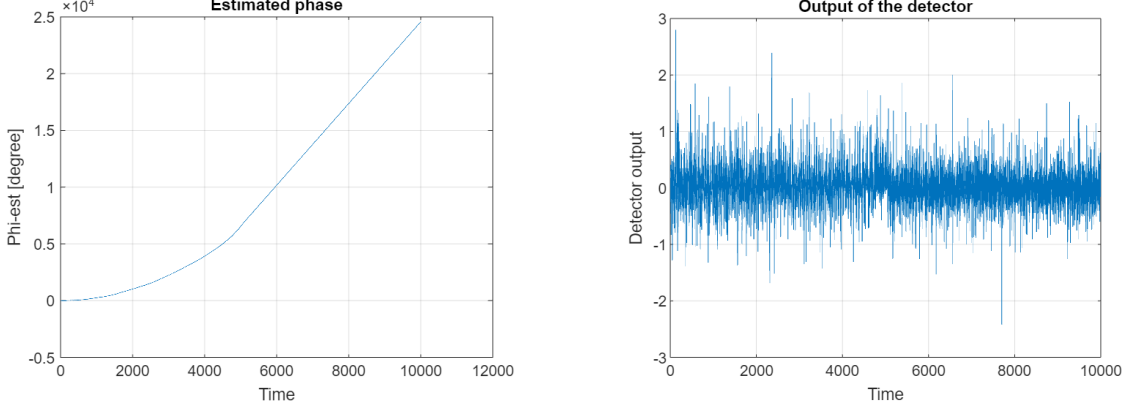


Figure 7: DPLL response to $\frac{\delta f}{R_s} = 1\%$; $\Delta\phi = 0^\circ$ for $E_b/N_0 = 7$ dB

1.2.2 Tracking

To achieve a phase-lock with minimized lock time, we assume a phase offset of zero degrees $\Delta\phi = 0^\circ$. The phase jitter has the expression:

$$\sigma_\theta^2 = E \left((\theta - \hat{\theta})^2 \right)$$

We plot in a logarithmic scale to show the property of the phase jitter $\sigma_\theta^2 \propto B_L T \frac{E_b}{N_0} \Leftrightarrow \log_{10}(\sigma_\theta^2) = \log_{10}(B_L T) - \log_{10} \left(\frac{E_b}{N_0} \right) + Cte$

The variance, σ_θ^2 , exhibits a direct proportionality to the loop bandwidth and time product over the bit energy to noise power spectral density ratio. When graphically depicted on a log scale, this relationship translates to a simple linear function: the logarithm of the variance against the system parameters is the difference of their respective logarithms, adjusted by a constant.

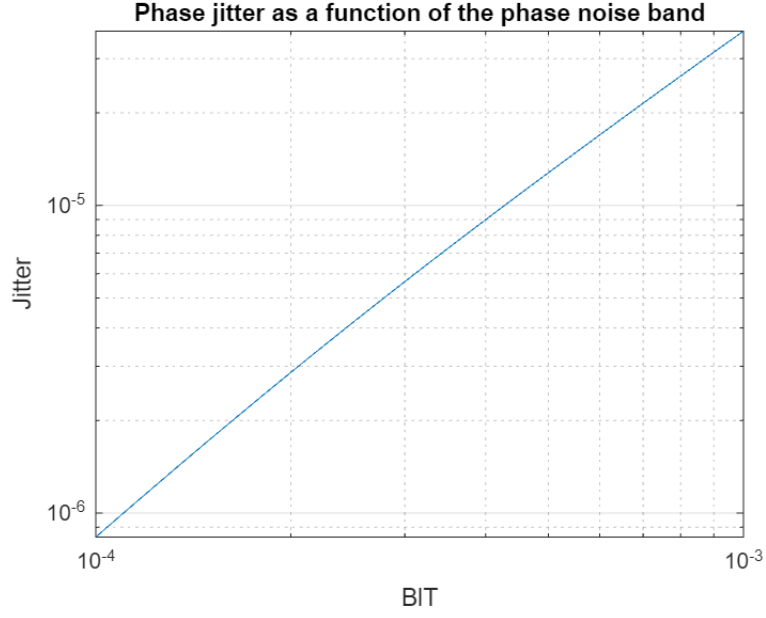


Figure 8: Phase jitter in function of the loop bandwidth for $\frac{E_b}{N_0} = 10dB$

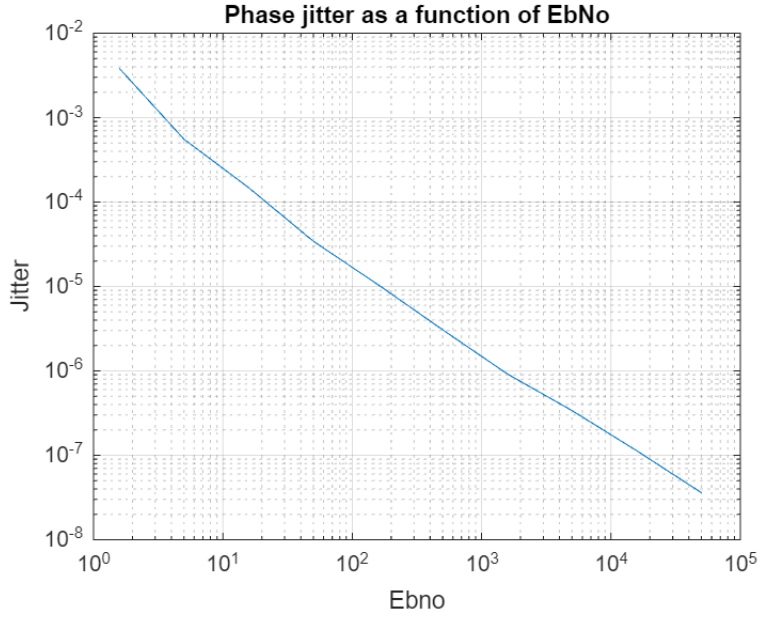


Figure 9: Phase jitter in function of the loop bandwidth for $B_L T = 10^{-2.5}$

2 QPSK modulation : DD PED

2.1 S-curve

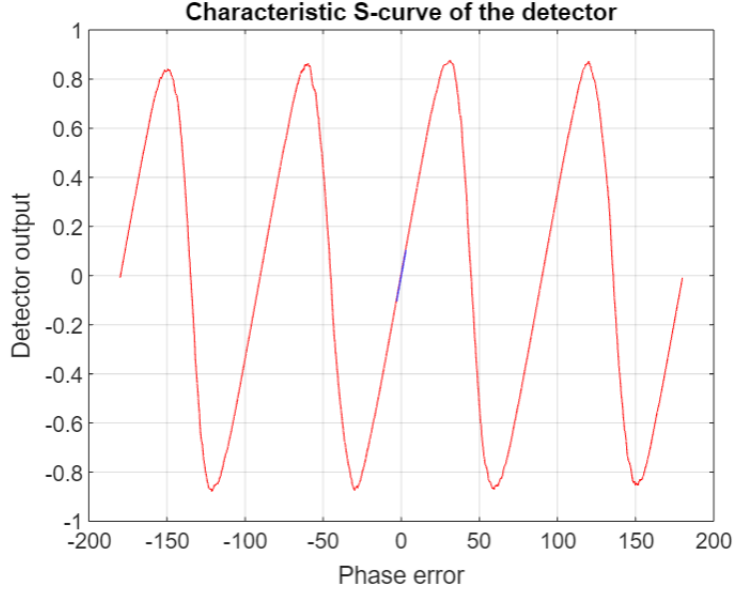


Figure 10: S-curve for a phase between -180° and 180°

2.1.1 Where does this expression comes from?

The likelihood function for phase estimation can be articulated as follows:

$$L(\hat{\theta}) = \exp \left(\sum_k \Re \left\{ d_k^* e^{-j\hat{\theta}} p(k, \hat{\tau}) \right\} \right)$$

where $\hat{\theta}$ represents the estimated phase, d_k^* is the complex conjugate of the received symbol, and $p(k, \hat{\tau})$ denotes the pilot symbol influenced by timing offset $\hat{\tau}$.

Differentiating the likelihood function with respect to $\hat{\theta}$, we obtain the gradient as:

$$\frac{dL(\hat{\theta})}{d\hat{\theta}} = \sum_k \text{Im} \left\{ d_k^* e^{-j\hat{\theta}} p(k, \hat{\tau}) \right\},$$

which simplifies to the imaginary part of the product of the received symbol and the pilot symbol after phase adjustment.

For a Decision-Directed (DD) implementation, the estimated symbol $\hat{d}_k$ is considered to be the complex conjugate of the received symbol, that is, $\hat{d}_k = d_k^*$. Therefore, the

expression for the phase error detector becomes:

$$u(k, \hat{\theta}) = \text{Im} \left\{ \hat{d}_k^* e^{-j\hat{\theta}} p(k, \hat{\tau}) \right\},$$

where $u(k, \hat{\theta})$ signifies the output of the phase error detector for the k -th symbol.

For a Data-Aided (DA) approach, the estimated symbol $\hat{d}_k$ aligns with the complex conjugate of the decision symbol, thereby: $\hat{d}_k = d_k^*$

2.1.2 DA (Data-Aided) implementation

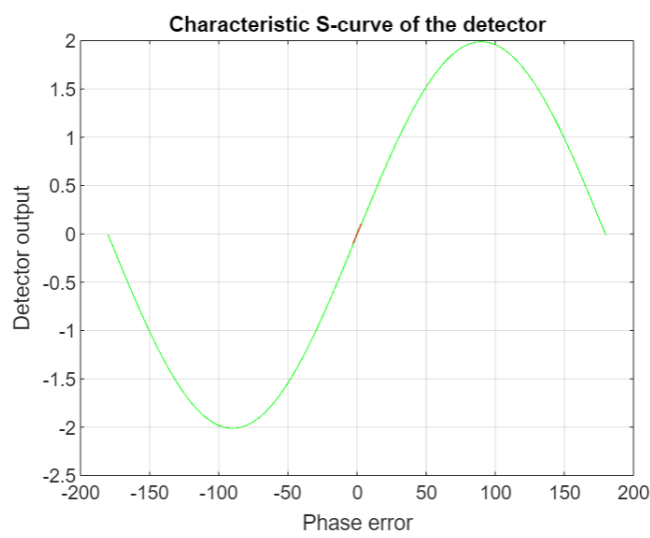


Figure 11: S-curve for $\frac{E_b}{N_0} = 5dB$

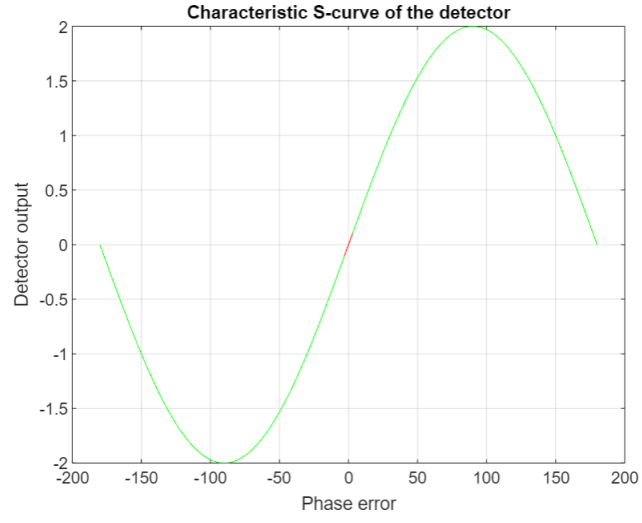


Figure 12: S-curve for $\frac{E_b}{N_0} = 10dB$

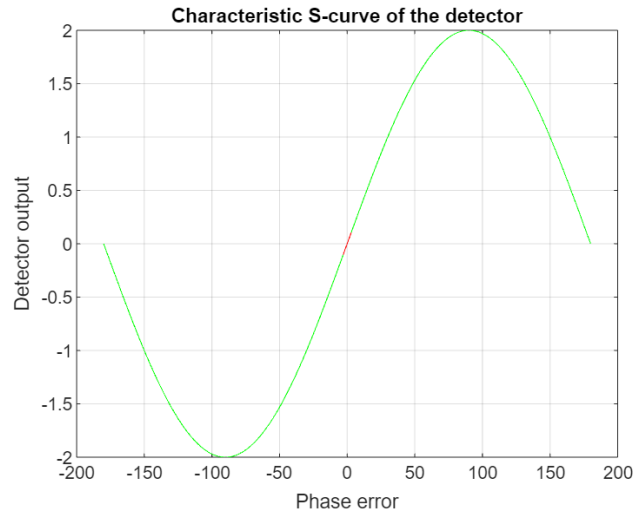


Figure 13: S-curve for $\frac{E_b}{N_0} = 100dB$

2.1.3 DD (Data-Aided) implementation

zrgqswgdfh<g

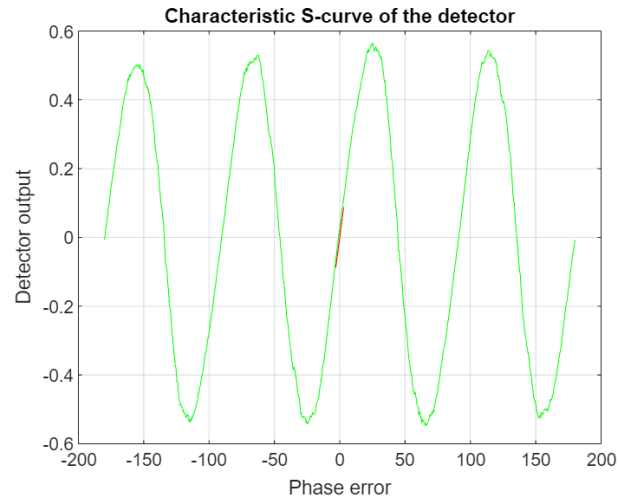


Figure 14: S-curve for $\frac{E_b}{N_0} = 10dB$

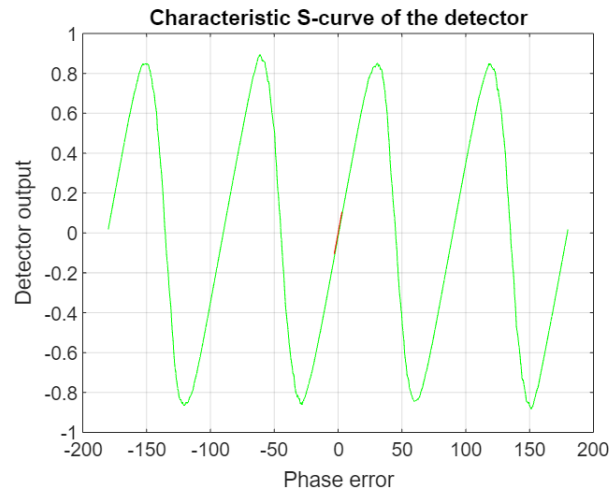


Figure 15: S-curve for $\frac{E_b}{N_0} = 10dB$

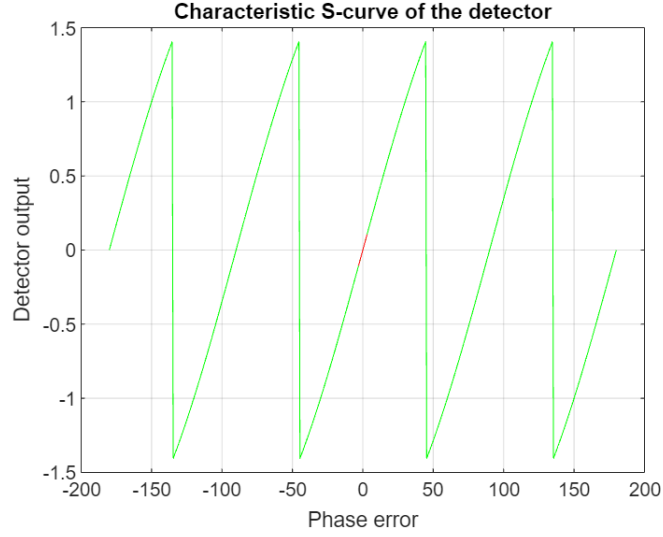


Figure 16: S-curve for $\frac{E_b}{N_0} = 100dB$

2.2 Closed loop

2.2.1 Adaptation to DD PED

To adapt our code on Matlab we change the following lines : `amsmath`

The equations corresponding to the MATLAB code are as follows:

$$recu_deph = \exp(-j \cdot phi_est[ii]) \cdot recu + noise; \quad (1)$$

$$symb_est = sign(real(recu_deph)) + j \cdot sign(imag(recu_deph)); \quad (2)$$

$$out_det[ii] = \frac{imag(conj(symb_est) \cdot recu_deph[ii])}{slope}; \quad (3)$$

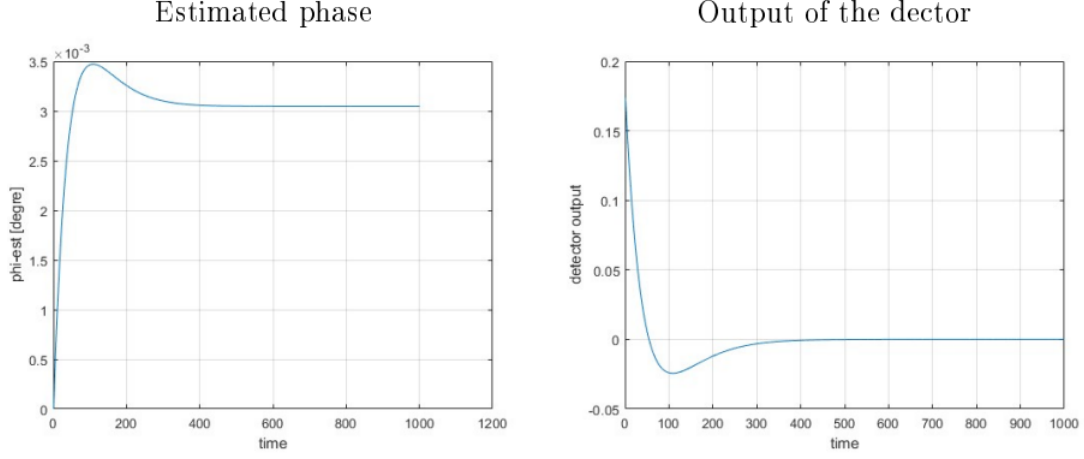


Figure 17: Phase shift $\Delta\phi = 10^\circ$ for $\frac{E_b}{N_0} = 100dB$

2.2.2 Phase shift test

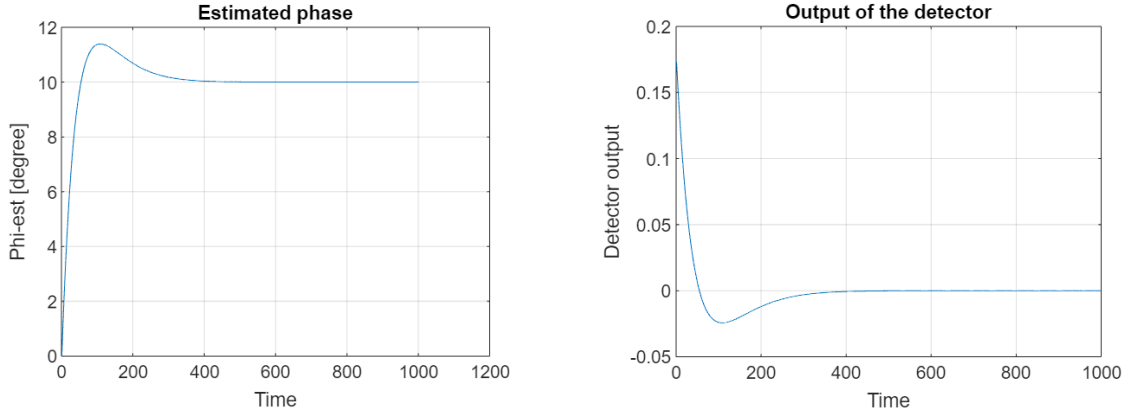


Figure 18: Phase shift $\Delta\phi = 10^\circ$ for $\frac{E_b}{N_0} = 100dB$

3 8PSK modulation : NDA & DD PED

3.1 S-curve

For non-differential 8PSK modulation, the expression is given by $u(k, \hat{\theta}) = \text{Im} \left\{ p(k, \hat{\theta}) e^{-j\theta} \right\}^8$. In the case of differential detection (DD), the expression remains unchanged from the previous scenario and is $u(k, \hat{\theta}) = \text{Im} \left\{ d^* k e^{-j\theta} p(k, \hat{\theta}) \right\}$.

3.1.1 NDA PED

In both Non-Differential Amplitude (NDA) and Differential Demodulation (DD), it is noteworthy that the phase ambiguity is $\frac{\pi}{8}$. This is observed as both curves exhibit zeros at points where $\tilde{\theta} - \theta = k\frac{\pi}{8}$.

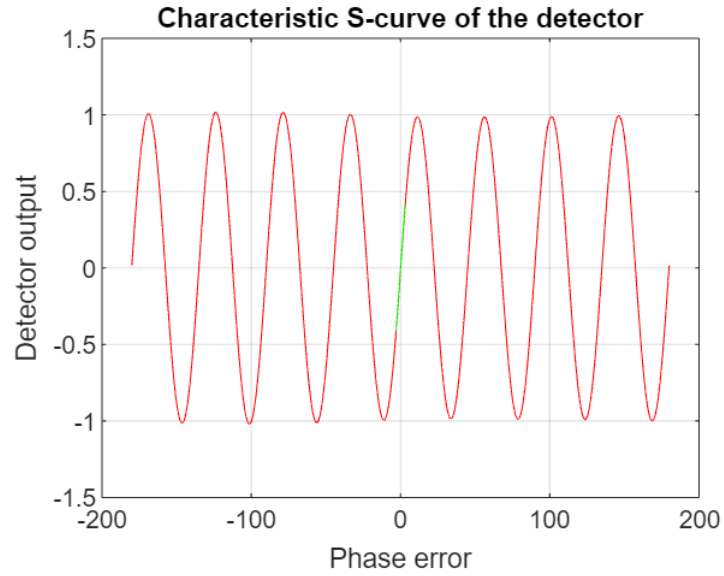


Figure 19: S-curve for $\frac{E_b}{N_0} = 100dB$

3.1.2 DD PED

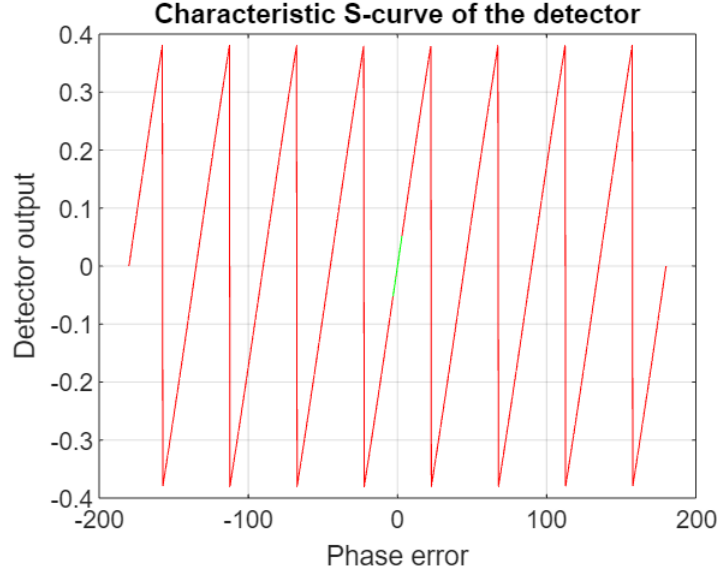


Figure 20: S-curve for $\frac{E_b}{N_0} = 100dB$

3.2 Closed loop

3.2.1 NDA PED

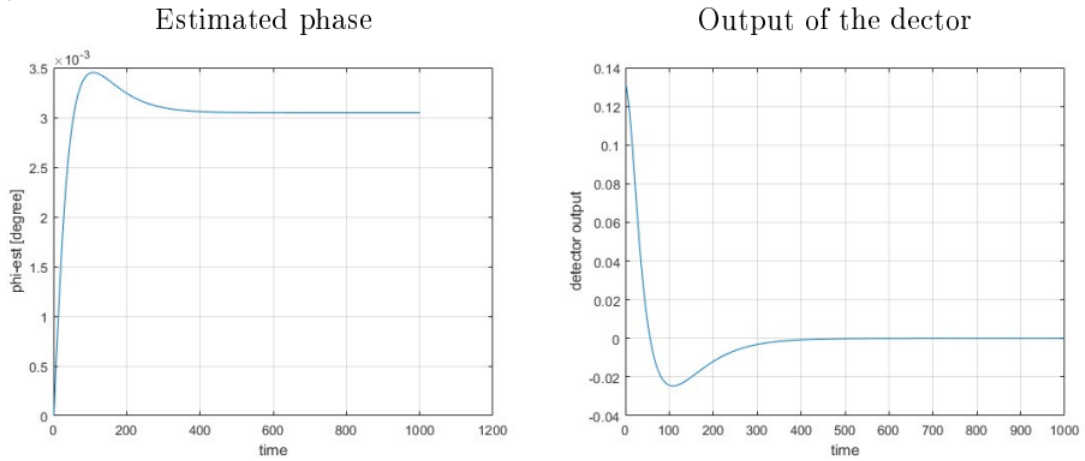


Figure 21: Phase shift $\Delta\phi = 10^\circ$ and $E_b/N_0 = 100$ dB

3.2.2 DD PED

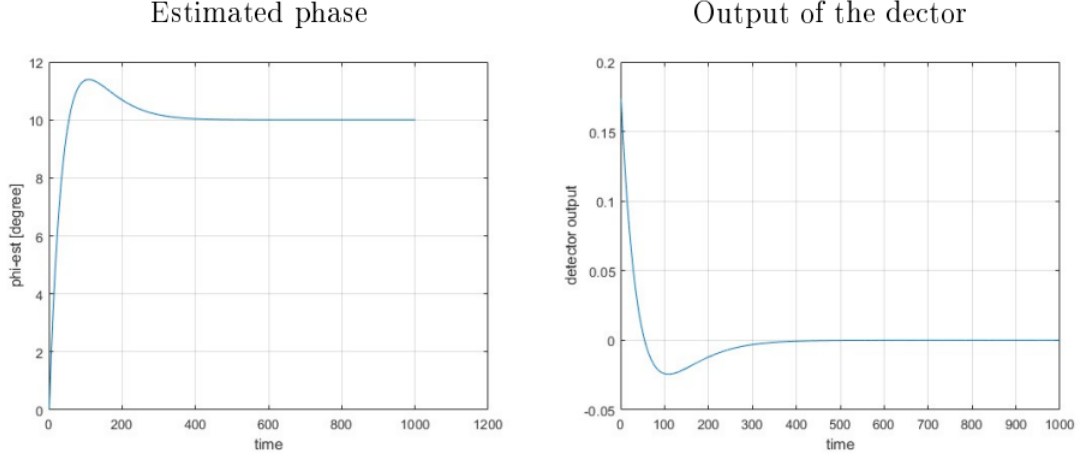


Figure 22: Phase shift $\Delta\phi = 10^\circ$ and $E_b/N_0 = 100$ dB

4 Conclusion

In this lab, we implemented and analyzed a Digital Phase-Locked Loop (DPLL) for QPSK and 8PSK modulations. We explored Non-Data Aided (NDA) and Data-Directed (DD) Phase Error Detectors (PEDs), investigating S-curves, phase ambiguity, and closed-loop performance.

For QPSK, we observed a phase ambiguity of $\frac{\pi}{2}$ in NDA. The DPLL demonstrated effective acquisition and tracking. Extension to 8PSK revealed a phase ambiguity of $\frac{\pi}{8}$. The DPLL exhibited robust performance in closed-loop scenarios, adapting to various conditions.

In conclusion, the implemented DPLL showcased its adaptability and efficiency in mitigating phase ambiguity and tracking signals. These findings underscore its applicability in communication systems.